Part of the problem is that there is no single "Copenhagen interpretation" and never really was. "The name 'Copenhagen interpretation' has gotten pretty slippery," said Nina Emery, a philosopher of physics at Mt. Holyoke. "The semantic confusion makes it easy for physicists to avoid dealing with those flaws directly. For instance, when you push them on the idea that measurements cause collapse… they shift and start talking about some kind of Bohrian view or about the [mathematics of the theory]. And if you point out the issues with those views (e.g. who knows what the former is; and the latter isn't even a complete interpretation), they go back to talking about [measurements causing collapse]." The flexibility afforded by these contradictory positions makes it easy to fend off any attacks on "the" Copenhagen interpretation. Physicists can simply hop from one position to another—sometimes without even realizing that they've done it.

But if you're an instrumentalist—if you think that science is merely a tool for predicting the outcomes of experiments and nothing more—then this kind of hopping around isn't a problem, because questions of interpretation are pointless and unscientific anyhow. It doesn't matter whether you hold a single consistent position about the meaning of quantum theory. All that matters is what you can directly observe. These kinds of positivist-like ideas are still very popular among physicists, especially when the subject of quantum physics comes up. Zeilinger has suggested that the "message of the quantum" is the fairly positivist idea that "the distinction between reality and our knowledge of reality, between reality and information, cannot be made." And the eminent physicist Freeman Dyson, like Rosenfeld before him, has suggested that it may be impossible to observe any consequences of a theory of quantum gravity, which he claimed "would imply that theories of quantum gravity are untestable and scientifically meaningless," in true positivist style.

Yet philosophers have known for more than half a century that the positivism underpinning statements like these is fundamentally flawed. And philosophers of physics today almost unanimously reject the Copenhagen interpretation. (Logical empiricism of a sort has made a comeback since 1980, but scientific realism is still the standard position among philosophers of physics—and even the most staunch defenders of empiricism today agree that the sort of naive positivism deployed in standard defenses of